



Molecular Docking

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Docking experiment

Simulation of protein-ligand (or protein-protein) binding

- **Should answer following questions:**
 - **Does ligand 1 bind to protein A?**
 - **How does it bind?**
 - **Which interactions are essential?**
 - **Which amino acid residues are important**



Docking Experiment

Areas of application:

– Identify binding mode of a ligand

- Understanding the interactions can lead to the design of more active compounds
- Suggestions for mutation studies

– Identify new compounds

- Virtual screening of large databases

Docking experiment

- Involved processes

Binding site recognition, identification and characterization
Find possible sites of interactions – binding sites, active sites



Placement of the ligand (protein)
Find possible orientations and interactions – generating docking poses



Scoring the docking pose
Evaluate/rate interactions, rank poses



Binding site recognition

Emil Fischer's lock-and-key model (1894)

- The ligand-protein binding is described by the interaction of 2 rigid bodies with Perfectly matching shapes





Binding site recognition

Motivation: 3D structures are available for proteins whose interactions with small molecules (ligands) are not known

Goal: Given a protein structure, predict its ligand bindings

Applications:

- Function prediction
- Drug discovery

Questions:

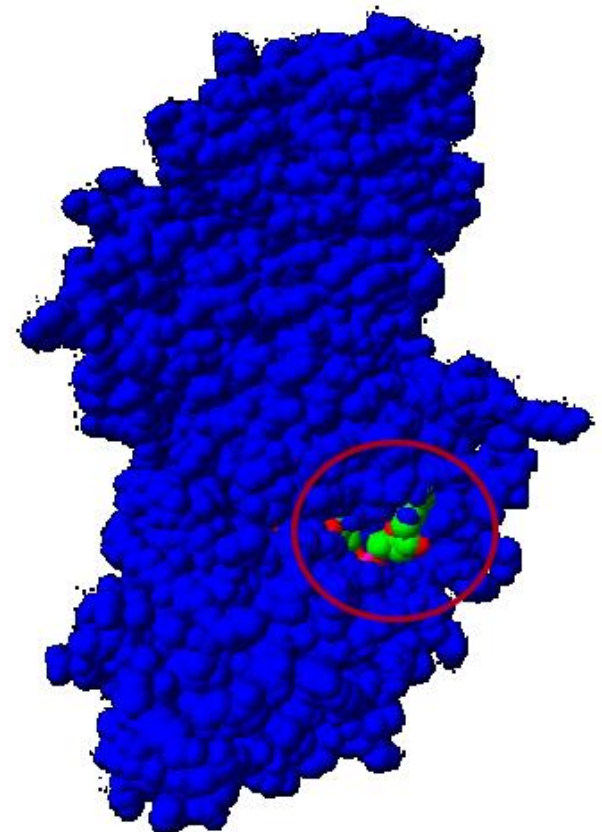
Where will the ligand bind?

Which ligand will bind?

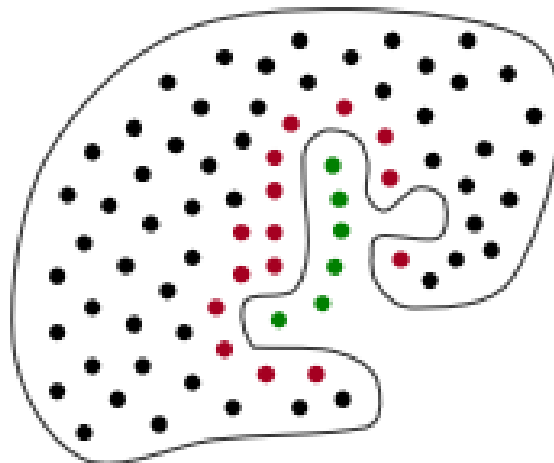
How will the ligand bind?

When?

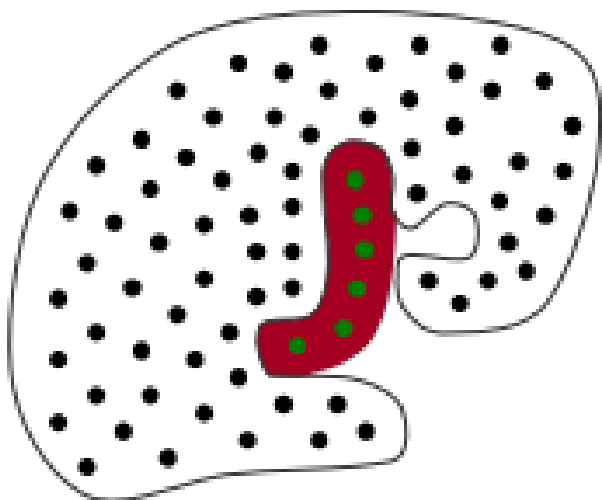
Why?



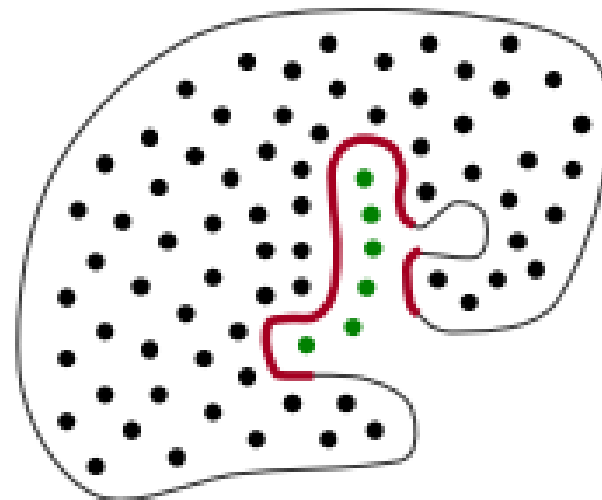
Binding Site Representations



Set of atoms/residues



Volume



Surface

Binding Site Representations

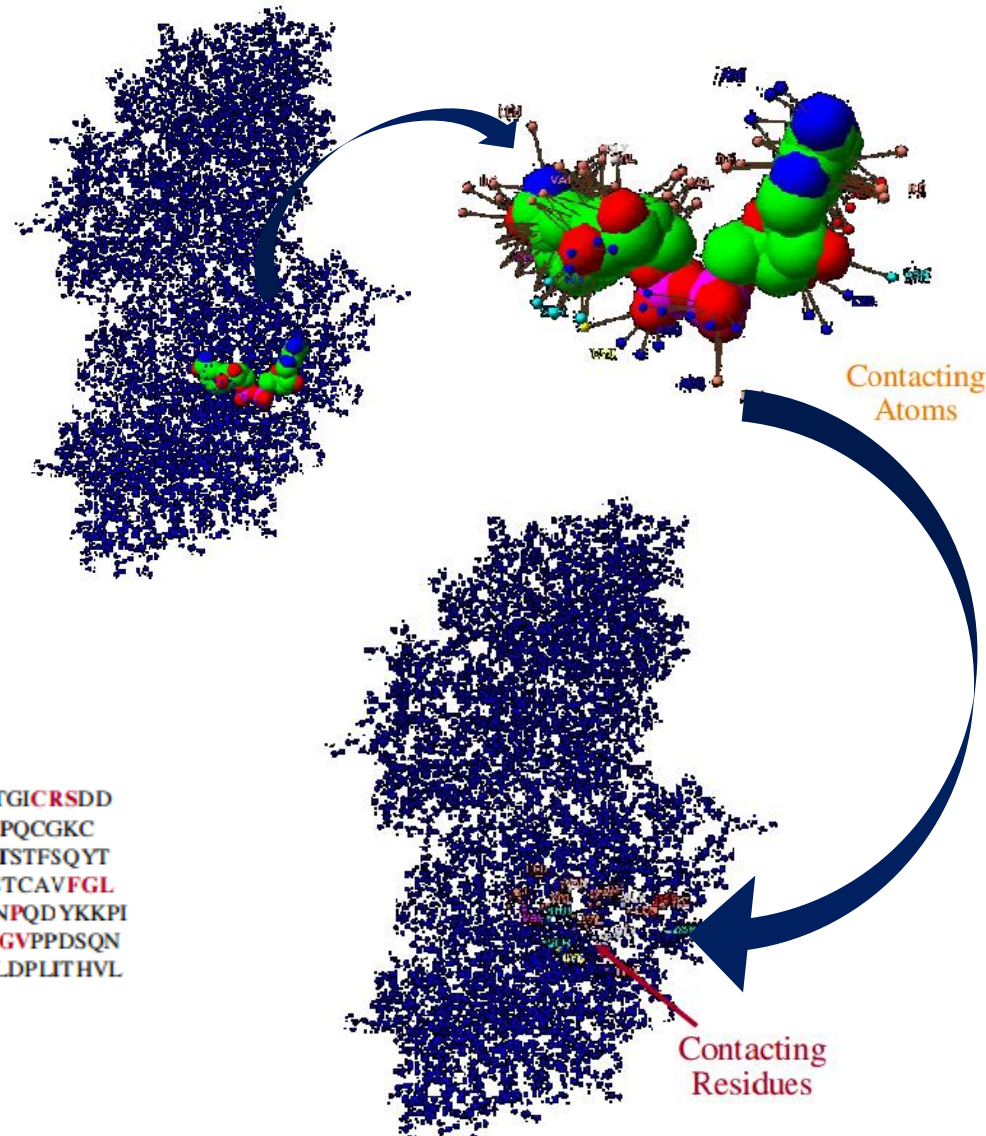
Set of atoms/residues

Sequence

STAGKVIKCKAAVLWEEKKPFSEIEVEVAPPKAHEVRIKMVATGICRSDD
HVVS GTLVTPLPVIAGHEAAGIVESIGEGVTTVRPGDKVIPLFTPQCGKC
RVCKHPEGNFCLKNDLSMPRGTMQDGTSRFTCRGKPIHHFLGTSTFSQYT
VVDEISVAKIDAASPLEKVCLIGCGFSTGYGS AVKVAKVTQGSTCAVFGL
GGVGLSVIMGCKAAGAARIIGVDINKDKFAKAKEVGATECVNPQDYKKPI
QEVLT EMSNGGVDFSFEVIGRLDTMVT ALSCCQEAYGVSIVGVPPDSQN
LSMNPMLLLSGRTWKGAIFGGFKSKDSVPKLVADFMKKFALDPLITHVL
PFEKINEGFDLLRSGESIRTILTF

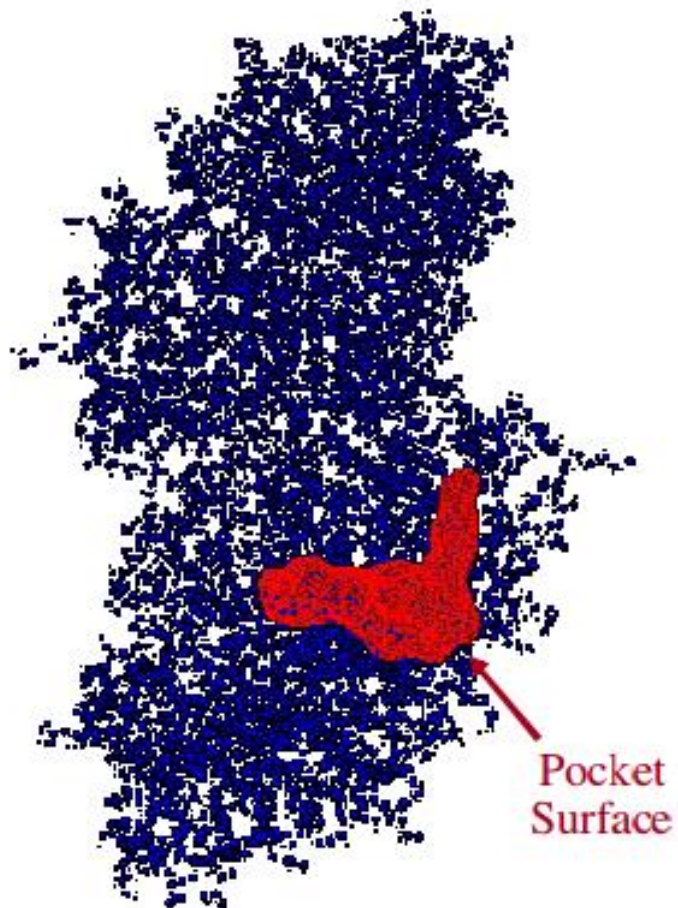
Sequence

STAGKVIKCKAAVLWEEKKPFSEIEVEVAPPKAHEVRIKMVATGIC**RSDD**
HVVS GTLVTPLPVIAGHEAAGIVESIGEGVTTVRPGDKVIPL**FTP**QCGKC
RVCKHPEGNFCLKNDLSMPRGTMQDGTSRFTCRGKPIHHFLGTSTFSQYT
VVDEISVAKIDAASPLEKVCLIG**CGFS****TGYGS** AVKVAKVTQGSTCAV**FGL**
GGVGLSVIMGCKAAGAARIIG**VDI****NKD****KFAKA**KEVGATECVNPQDYKKPI
QEVLT EMSNGGVDFSFE**VIGRLD****TMVT** ALSCCQEAYGVSIV**GV**PPDSQN
LSMNPMLLLSGRTWKGA**IFG**FKSKDSVPKLVADFMKKFALDPLITHVL
PFEKINEGFD**LLRSGESIR**TILTF

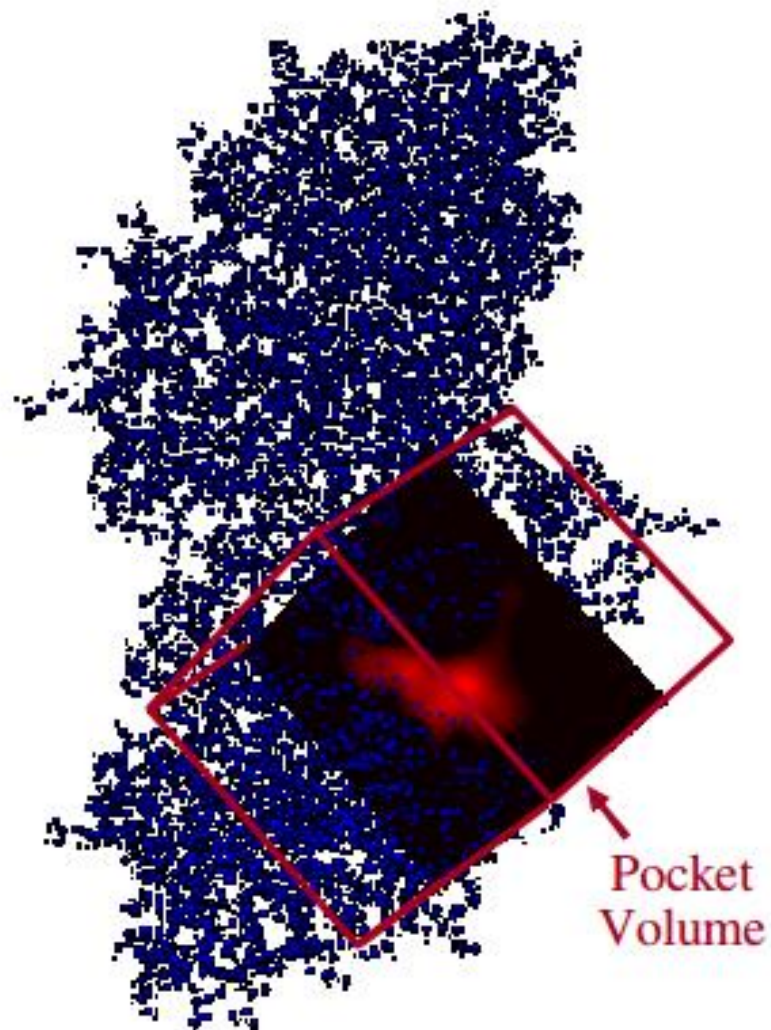


Binding Site Representations

Surface



Volume



Binding site recognition

Shape complementarity

- lock-and-key model

basis of most binding site identification methods

- BUT: proteins and ligands are not rigid, but dynamic objects
- softening of the model!

Physico---chemical complementarity

- van-der-Waals interactions
- electrostatic interactions
- hydrogen-bonds
- hydrophobic interactions
- solvent interactions

binding pose should be energetically favorable!!!

Geometric approaches

Grid-based

Cover the protein into a 3D grid,
Empty grid points are then defined a pockets if they satisfy a number of
geometric or energetic conditions

Sphere-based

A set of probe spheres are placed on protein surface
Pocket spheres are those generated probe spheres that satisfy a number of
geometric conditions among the generated probe spheres

α -shape based

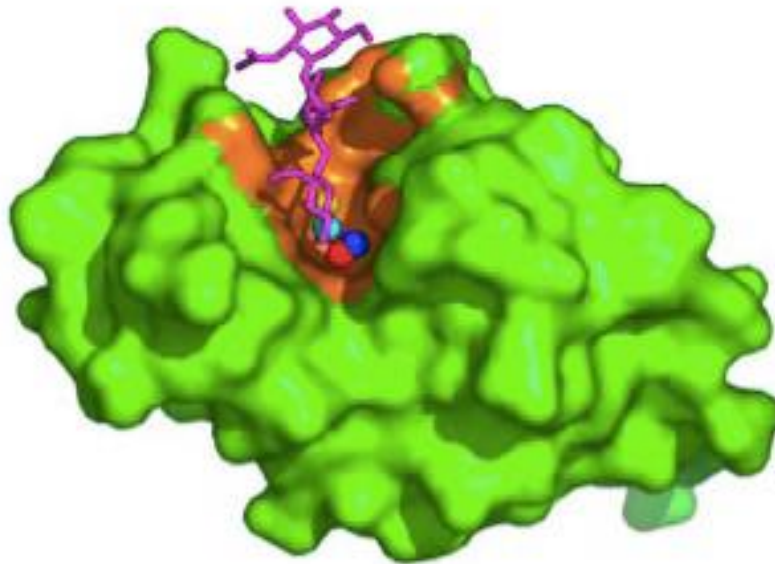
Is defined as a subset of Delaunay triangulation of protein atoms, omitting edges
longer than the sum of the radii of two atoms

Methods for binding site identification

Geometric (Grid) approaches:

- detect cavities and crevices on the protein surface

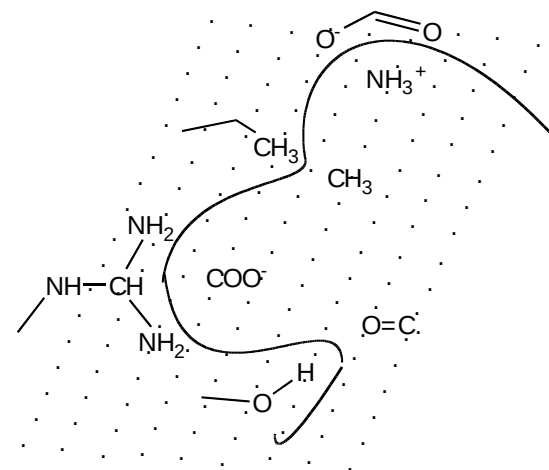
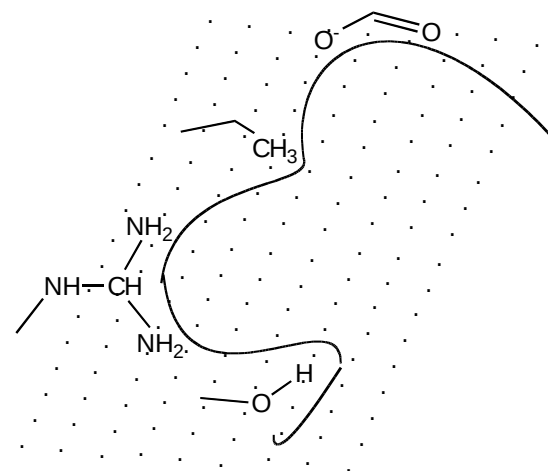
Software: POCKET, LIGSITE, SURFNET, CAST, PASS



Grid

- a) Prepare a 3D lattice of grid points encompassing the binding site
- b) Determine interaction energy of different types of functional groups with binding site at each grid point.
- c) Select favorable interaction sites
- d) Determine relative spatial orientations of the selected interaction sites

Types of functional groups and their positions yields a pharmacophore. Grid may also be used to calculate interaction energies of ligands with a receptor.



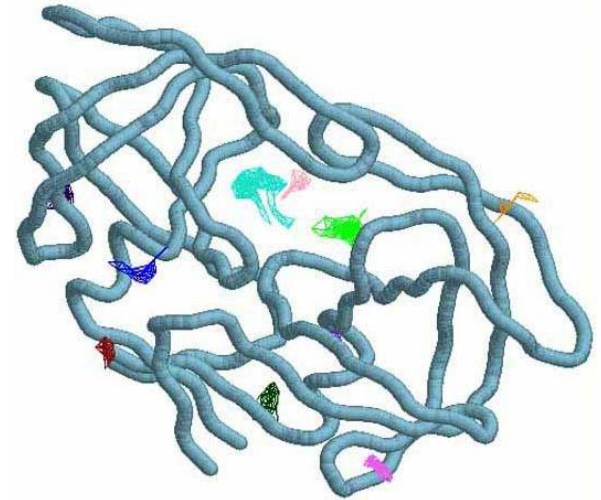
Q-Site Finder

Q-Site Finder locates ligand binding sites by clustering favorable regions for van der Waals (CH₃) probes on the protein surface

It uses the GRID forcefield parameters to estimate the interaction energies of probes placed at all points on a three dimensional grid that encompasses the entire protein.

Probes with favorable interaction energies are retained and are clustered. The clusters are ranked according to their total interaction energy.

The algorithm was shown to have a 90% success rate in the top three predicted sites when tested on 134 protein ligand complexes corresponding described by Nissink *et al.*



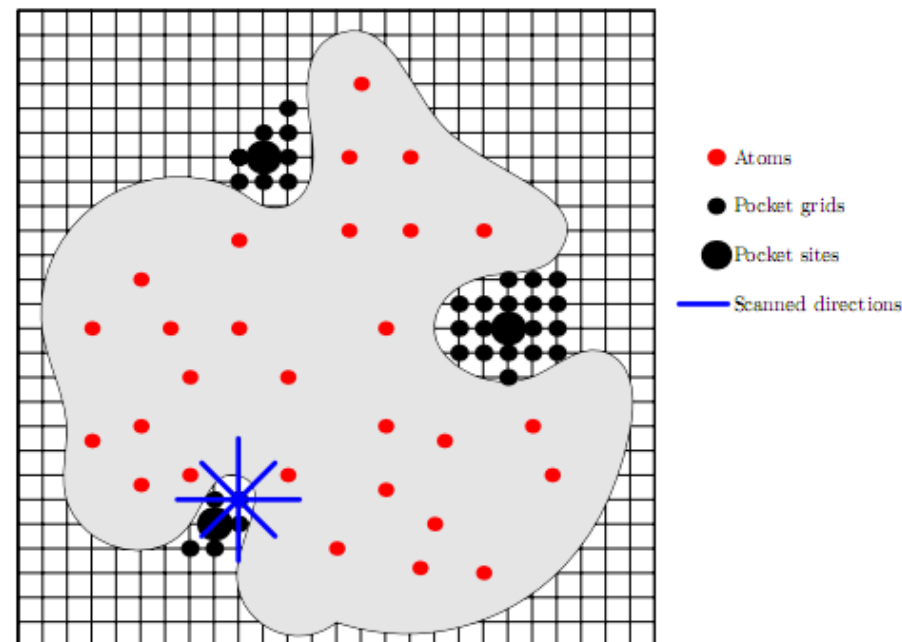
POCKET

1. Protein is projected onto a 3D-grid
2. They focused on PSP (protein-solvent-protein) events of the grids

When a straight line drawn from a grid point is enclosed on both side by protein atoms, the arrangement of the line (x, y and z direction) for that grid point is termed a PSP event

3. Consider Grid points within 3A of protein atom: protein (otherwise: solvent)
4. Grid points having more than a threshold number of PSP events are defined as pockets

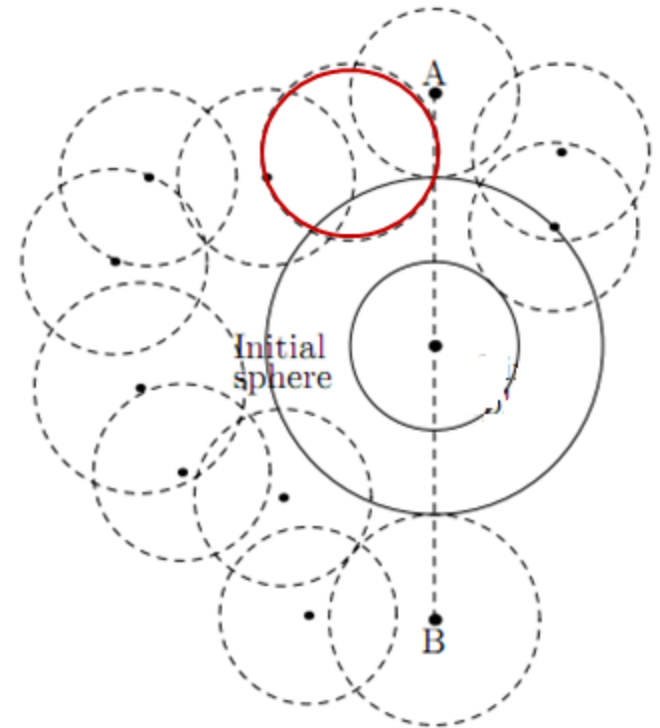
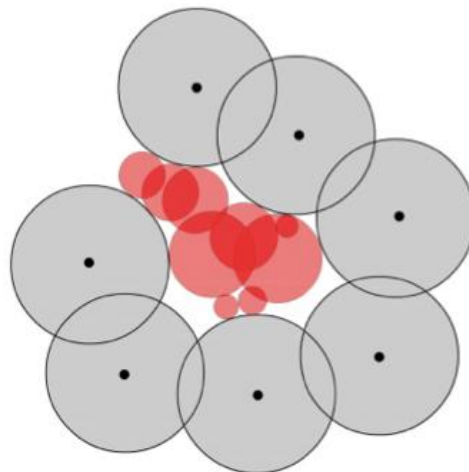
LIGSITE, LIGSITEcs, Pocket Picker



Sphere based

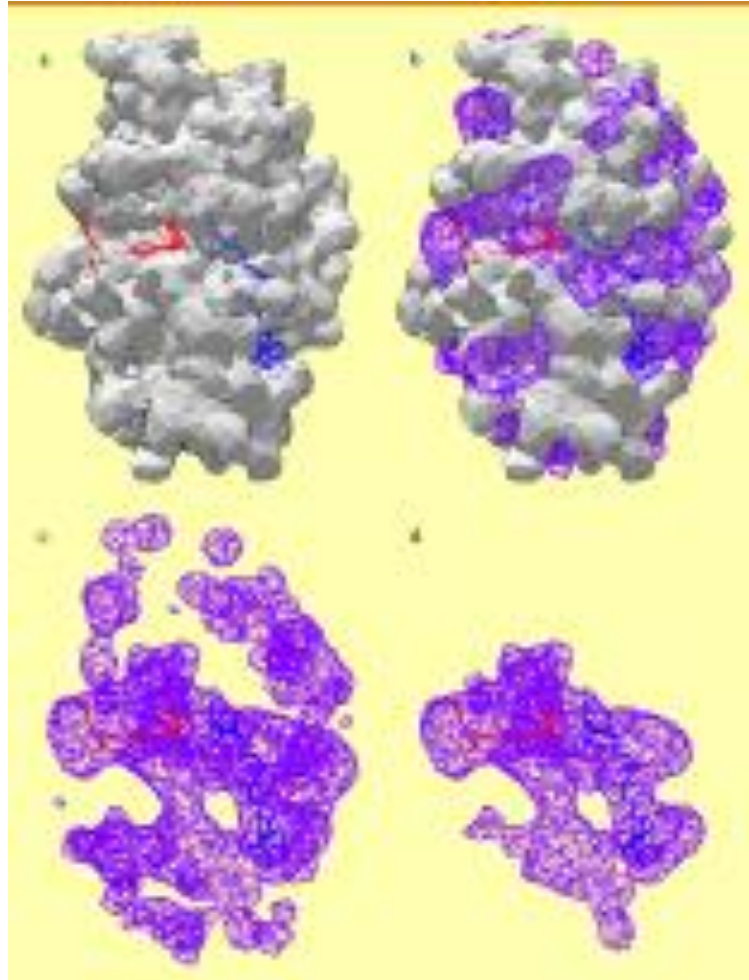
1. Generate sphere (called gap spheres) between two protein atoms → initial sphere
2. If the sphere contains any other atoms, reduce its radius until it just touches one protein atom → final sphere
3. Arrangement of final gap spheres describes pocket

SURFNET



Sphere based

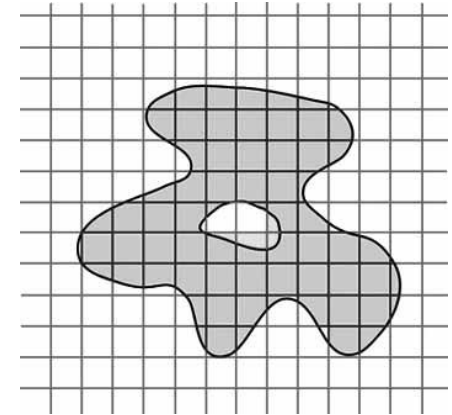
SURFNET



Delaney

The protein is placed within a three dimensional grid.

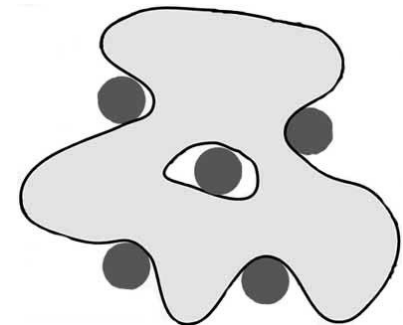
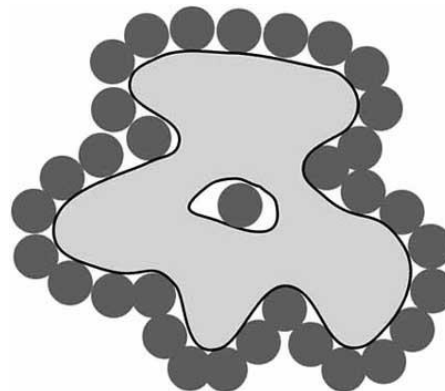
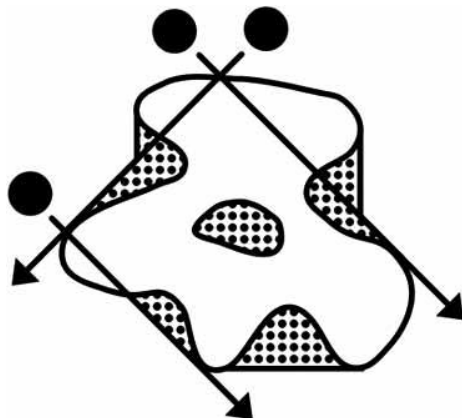
Where grid points intersect the protein, they are set to 'true', otherwise they are set to 'false'



The protein surface are defined to be grid points set to 'true' that are adjacent to grid points set to 'false'.

A monolayer of spheres is then added to the protein surface and the true/false representation is recalculated to redefine the surface.

A surface contraction then takes place, where a monolayer of particles/spheres is removed.

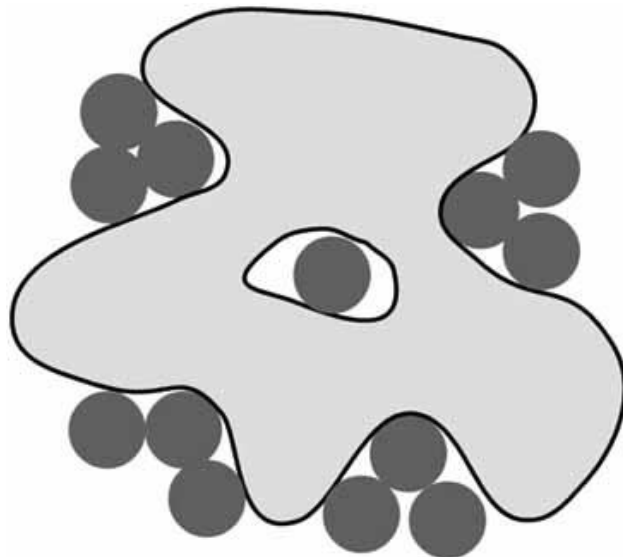


Delaney

Expansion into pockets can add particles that are not subsequently defined to be part of the protein surface

Some of the particles added to pockets survive the surface contraction. After repeated expansions and contractions (usually five to ten), the protein cavities are filled with particles

The particle count predict the potential binding site

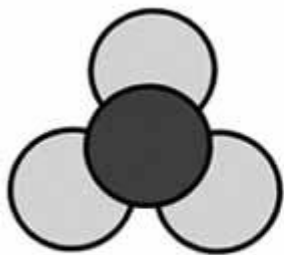


PASS

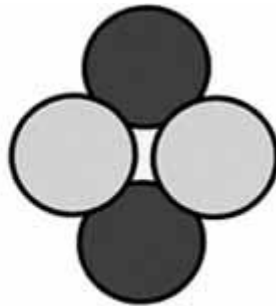
PASS uses a similar concept to Delaney. However, a different type of analysis is used to achieve a similar effect

The algorithm looks at all possible combinations of **three protein atoms**. If the three atoms are **close** enough together, the algorithm calculates the **two possible positions for a probe sphere** to just **touch the surface** of all three protein atoms.

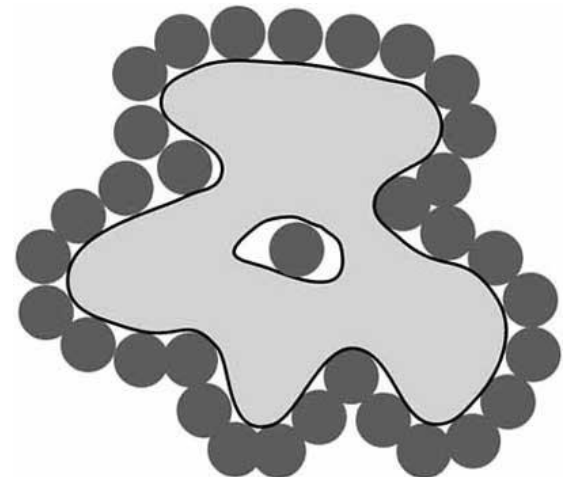
The probes are **rejected** if they **clash** with protein atoms. This results in a covering of the protein surface with probes, similar to the surface expansion of Delaney



From above



From the side



PASS (Filtration)

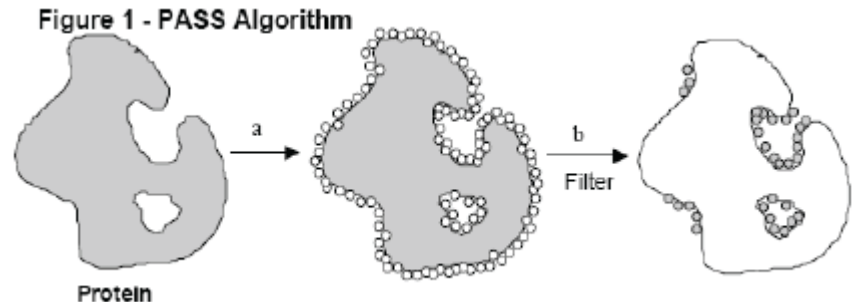
The **burial count** of each probe is measured by calculating the **number of protein atoms** found within an 8Å of the probe.

Probes in **pockets** have a **higher burial count** than those **outside** pockets.

A **burial count threshold** is applied to remove probes outside pockets. This filtration step leaves probes in pockets and cavities

Repeated cycles of addition of probe spheres followed by filtration to remove spheres not found in protein pockets causes accumulation of spheres in a similar fashion as in Delany

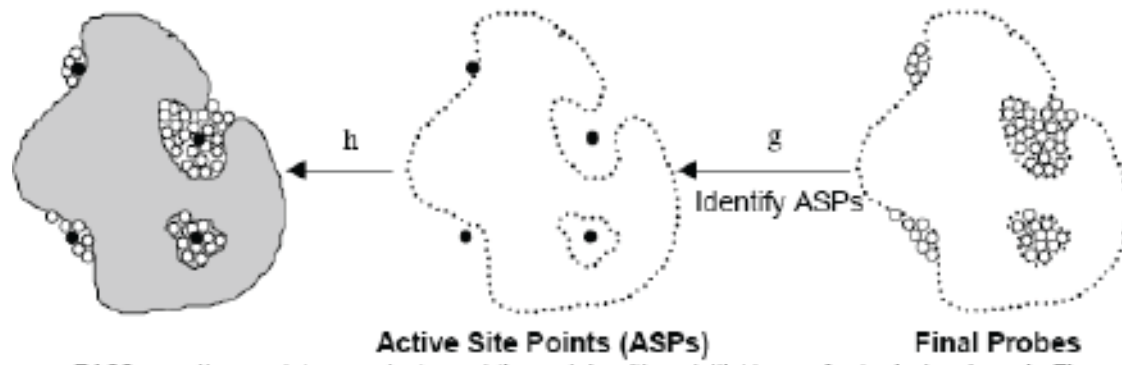
When these cycles of addition and filtration no longer lead to a change in the number of probes bound to the protein, the end point is reached



PASS

Final probes are weighted according to their burial degree and active Site points (ASPs) are defined ($\rightarrow$ central points)

Spheres with a certain burial degree around ASPs are kept as active binding site points



α -shape based

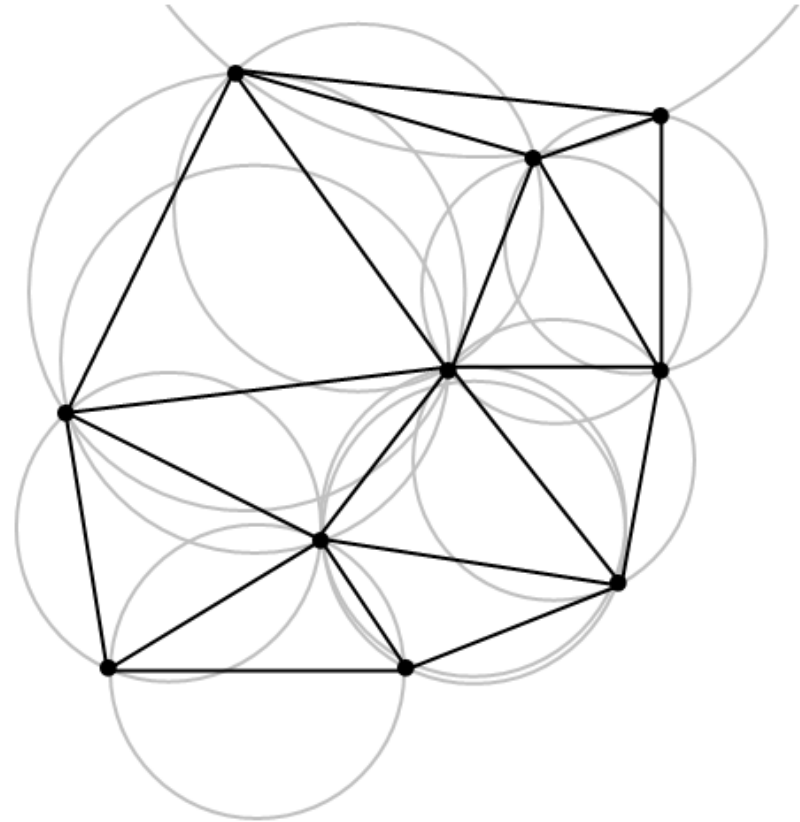
The α -shape is defined by configuring the space with a sphere with a radius α when the sphere does not contain any input point

When the sphere touches two points, they are connected by an edge.

When the sphere touches three points, a triangular face is defined by the points.

When the sphere touches only one point, the point is left as a singleton.

These points, edges and triangular faces together define the α -shape for the given point set.



α -shape based

Voronoi decomposition: is a way of dividing space into a number of regions

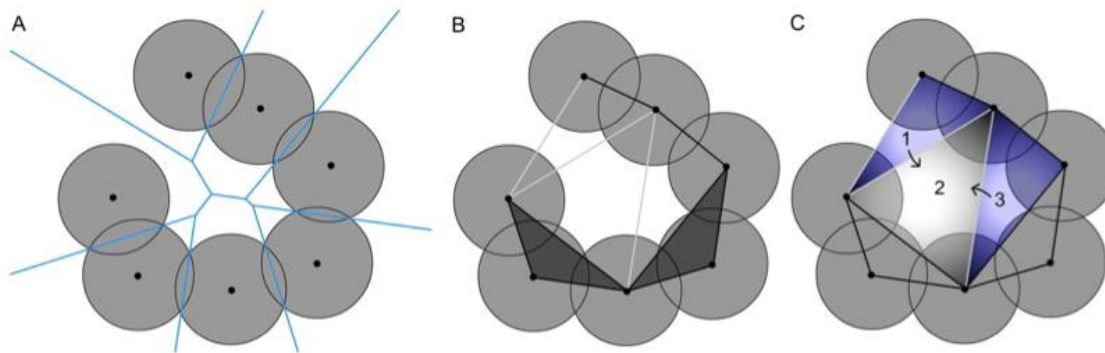
Delaunay triangulation: A Delaunay triangulation for a set P of points in a plane is a triangulation $DT(P)$ such that no point in P is inside the circumcircle of any triangle in $DT(P)$.

CAST

A. Voronoi decomposition is calculated from protein atoms

B. Delaunay triangulation is obtained from voronoi cells

C. „empty“ triangles are combined to describe the pocket



SURFNET: <http://www.biochem.ucl.ac.uk/~roman/surfnet/surfnet.html>

PASS: <http://www.ccl.net/cca/software/UNIX/pass/overview.shtml>

CAS: <http://cast.engr.uic.edu/cast>

Q- Site Finder: <http://www.bioinformatics.leeds.ac.uk/qsitefinder>